POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the functional view of the computer system with reference to each component. 8
   b) How data can be transferred between registers? Explain in detail. 7
2. a) Describe the arithmetic and logic circuit with its appropriate block diagram. 7
   b) Multiply $(14)_{10}$ and $(3)_{10}$ using unsigned binary multiplication method with data flow diagram. 8
3. a) Perform (-5) *(-3) using the Booth multiplication algorithm. 8
   b) Compare single and two address field addressing techniques used in micro-programmed implementation of control unit. 7
4. a) Does hardwired implementation of Control Unit makes control unit fast or slow explain? Mention disadvantages of Hardwired implementation. 8
   b) What is associative memory? How match logic works for associative memory? 7
5. a) Define DMA. Explain DMA controller with different DMA transfer modes. 7
   b) Explain the advantages of pipelining in the multiprocessor system. Describe the instruction pipeline with examples. 8
6. a) Why cache coherency occur in multiprocessor system? What are the various ways to troubleshoot the problem? 8
   b) Explain Flynn's classification of computer system. 7
7. Write short notes on: (Any two) 2×5
   a) Optical disk
   b) RISC Vs CISC
   c) Multi Core Organization

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